

1. Spontaneous Process

自发的

enthalpy 焓 entropy 熵

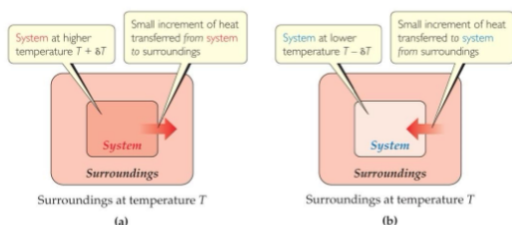
- Processes that are spontaneous in one direction are **nonspontaneous** in the reverse direction.

在外界环境不动情
况下定义

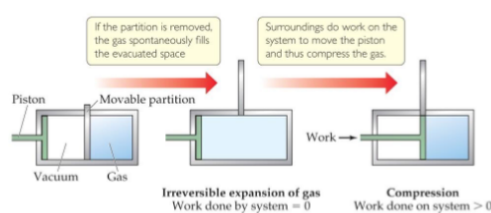
- Temperature and pressure can affect spontaneity.

→ 水结冰, 冰融化 (T 控制)

2. Reversible and irreversible processes



Reversible process: The system changes so that the system and surroundings can be returned to the original state by exactly reversing the process. This maximizes work done by a system on the surroundings.



Irreversible processes cannot be undone by exactly reversing the change to the system or cannot have the process exactly followed in reverse. **Any spontaneous process is irreversible!**

Chemical

3. Entropy → randomness of a system

$$\Delta S = S_{\text{final}} - S_{\text{initial}} \quad \text{state function} \quad S \text{ 本身也是}$$

$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{固定 } T)$$

→ phase change

Second law $(\Delta S_{\text{surr}} = \frac{-q_{\text{sys}}}{T_{\text{isothermal}}})$

reconcile
调和, 调解

Second Law of Thermodynamics

- The **entropy of the universe** increases in any spontaneous processes.

- This results in the following relationships:

Reversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$

Irreversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$

恒压 $\Delta S_{\text{surr}} = \frac{-\Delta H_{\text{sys}}}{T}$

Entropy on the Molecular Scale

- Microstate: A single possible arrangement of **position** and **kinetic energy** of molecules

↓
↓
Boltzman

Chemical
Thermodynamics

$$S = k \ln W$$

The number of microstates and, therefore, the entropy tend to increase with increases in

- temperature.
- volume.
- the number of independently moving molecules.



Entropy and Physical States

- Entropy increases with the freedom of motion of molecules.
- $S(g) > S(l) > S(s)$
- Entropy of a system increases for processes where
 - gases form from either solids or liquids.
 - liquids or solutions form from solids.
 - the number of gas molecules increases during a chemical reaction.

Increasing entropy

- The entropy of a pure crystalline substance at absolute zero is 0.
- Consider all atoms or molecules in the perfect lattice at 0 K; there will only be *one* microstate.
- $S = k \ln W = k \ln 1 = 0$

standard Entropies

at 298 K 1 atm

Standard Entropies

- The reference for entropy is 0 K, so the values for elements are *not* 0 J/mol K at 298 K.
- Standard molar enthalpy for gases are generally greater than liquids and solids. (Be careful of size!)
- Standard entropies increase with molar mass.
- Standard entropies increase with number of atoms in a formula.

Table 19.1 Standard Molar Entropies of Selected Substances at 298 K

Substance	S° (J/mol·K)
H ₂ (g)	130.6
N ₂ (g)	191.5
O ₂ (g)	205.0
H ₂ O(g)	188.8
NH ₃ (g)	192.5
CH ₃ OH(g)	237.6
C ₆ H ₆ (g)	269.2
H ₂ O(l)	69.9
CH ₃ OH(l)	126.8
C ₆ H ₆ (l)	172.8
Li(s)	29.1
Na(s)	51.4
K(s)	64.7
Fe(s)	27.23
FeCl ₃ (s)	142.3
NaCl(s)	72.3

Chemical
Thermodynamics

4. Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

- If ΔG is negative, the forward reaction is spontaneous ($\Delta S_{\text{universe}}$ **+**).
- If ΔG is 0, the system is at equilibrium ($\Delta S_{\text{universe}}$ **0**).
- If ΔG is positive ($\Delta S_{\text{universe}}$ **-**), the reaction is spontaneous in the reverse direction.

Standard formation
生成

ΔG_f° 与 ΔH_f° 定义类似

$$\Delta G_{\text{rxn}}^\circ = \sum n \Delta G_f^\circ (\text{products}) - \sum m \Delta G_f^\circ (\text{reactants})$$

Reactants

$$\Delta G = \Delta G^\circ + RT \ln Q$$

如果标准状态, $Q=1$ $\Delta G = \Delta G^\circ$

如果平衡 $\Delta G = 0 \Rightarrow \Delta G^\circ = -RT \ln K$

B) thermodynamically favorable, because $K_3 > 1$